

E.3 / PAPER 1 / QUESTION 3

4 November 2021 / TZ0 / HL Paper 1 / original Q21 / 8821-6501

3. The mass of a nucleus of iron-56 (${}^{56}_{26}\text{Fe}$) is M .

What is the mass defect of the nucleus of iron-56?

- A. $M - 26m_p - 56m_n$
- B. $26m_p + 30m_n - M$
- C. $M - 26m_p - 56m_n - 26m_e$
- D. $26m_p + 30m_n + 26m_e - M$